



S I M A T S
ENGINEERING



SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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Real-Time Big Data Ingestion Pipeline from Distributed Sources

CSA1522-CLOUD COMPUTING & BIG DATA ANALYTICS

Guided by :

Dr. Rajaram P

Name of the student :

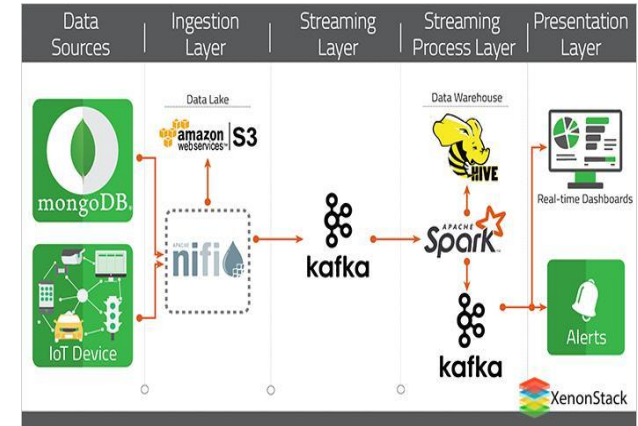
Gali Venkata Niranjan Reddy (192525199)

Mamidi Jaya Krishna (192524500)

Manukondu Sandeep (192525095)

Introduction

- Big Data applications generate massive volumes of data from multiple distributed sources in real time.
- Data may come from IoT devices, sensors, web applications, social media, servers, and enterprise systems.
- Traditional data processing methods may not handle high-velocity and continuously generated data efficiently.
- A real-time big data ingestion pipeline collects, processes, stores, and visualizes data with minimum delay.



Abstract

- Ingests high-velocity data from multiple sources.
- Uses **Apache Kafka** for real-time data ingestion.
- Uses **Apache Spark Streaming** for data processing.
- Uses **HDFS/S3** for large-scale data storage.
- Uses **Grafana/Kibana** for real-time visualization.
- Provides reliable and scalable data processing.

Literature Review

Author(s) & year	Study focus	Methodology	Key findings	Gaps identified
Springer KAIS study (2022)	Distributed real-time ETL for unstructured data	Novel architecture with stream-disk join	Proposes a partition-based, best-effort in-memory join technique that ingests unstructured multi-source streams without prior schema knowledge, cutting disk access.	Governance and schema-drift handling left for future work
Yaghi (2025)	Big data ingestion and preparation tools	Comparative review of tools	Analyzes Kafka, NiFi, Flume, Sqoop, and Spark across batch and stream ingestion, recommending separation of storage and processing layers.	Real-time governance and integration complexity remain unresolved
Beevi (2025)	Scalable data pipelines for real-time analytics	Comparative survey of streaming frameworks	Compares state-of-the-art frameworks on performance, fault tolerance, and cost-effectiveness for scalable ingestion pipelines.	Cost-effectiveness metrics not validated against production deployments
MDPI review (2025)	Evolution and challenges of real-time big data	Systematic review (2020–2025 literature)	Finds Kafka Streams, Flink, and Spark Streaming have cut latency and raised throughput; notes growing use of explainable AI and incremental learning atop ingestion pipelines.	Interoperability between edge computing and centralized streaming layers underexplored

Problem statements

- Modern applications generate huge volumes of data continuously from distributed sources such as IoT devices, applications, APIs, databases, and system logs.
- Traditional batch-processing systems cannot process this continuously generated data with low latency.
- Data arrives at different speeds and in different formats, making integration and processing difficult.
- Delayed data processing can affect real-time monitoring and decision-making.

Objectives

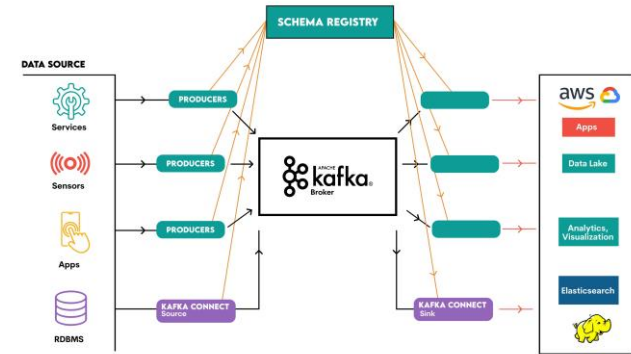
- To design a real-time pipeline for ingesting high-velocity data from multiple distributed sources.
- To use Apache Kafka for reliable distributed messaging and real-time data ingestion.
- To process and transform streaming data using Apache Spark Streaming.
- To store large-scale processed data using HDFS or cloud storage such as Amazon S3.
- To visualize real-time analytics and pipeline performance using Grafana/Kibana.

Module 1 – Multi-Source Data Ingestion Module

- Collects real-time data from multiple distributed sources such as IoT devices, sensors, applications, and server logs.
- **Apache Kafka** acts as the distributed messaging and data ingestion platform.
- Kafka topics organize incoming data streams and allow multiple consumers to process data independently.

Features

- Multi-Source Data Collection
- Apache Kafka Producers
- Kafka Topics and Partitions

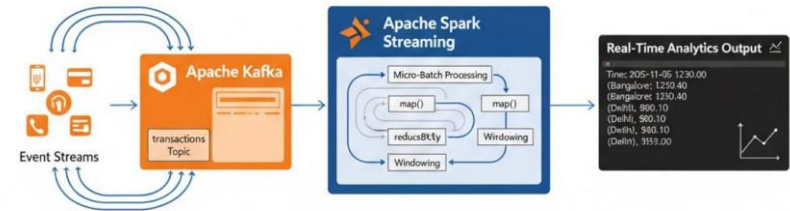


Module 2 – Stream Processing & Transformation Engine

- Receives continuous data streams from Apache Kafka.
- **Apache Spark Streaming** processes incoming data in real time or near real time.
- Performs data cleaning, filtering, transformation, aggregation, and feature extraction.

Features

- Real-Time Stream Processing
- Data Cleaning
- Data Filtering
- Data Transformation

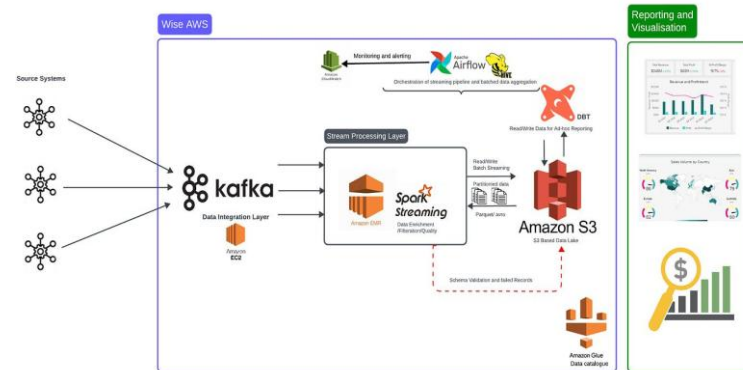


Module 3 – Cloud Storage & Visualization Dashboard

- Stores raw and processed big data in distributed storage systems.
- **HDFS** can be used for distributed on-premise storage, while **Amazon S3** provides scalable cloud-based object storage.
- Processed data is made available for analytics and long-term storage.

Features

- HDFS/S3 Data Storage
- Scalable Cloud Storage
- Historical Data Management
- Real-Time Data Visualization



Implementation

REAL-TIME BIG DATA PIPELINE

Processing Batch: 20/20

MODULE 1: DATA INGESTION

IoT Records: 10
Web Application Records: 10
Transaction Records: 10
Total Records Received: 30

MODULE 2: STREAM PROCESSING

Records After Processing: 10

MODULE 3: CLOUD STORAGE

Batch Stored Successfully
Location: /content/cloud_storage/processed_batch_20.csv

REAL-TIME VISUALIZATION DASHBOARD

RECORDS BY DATA SOURCE:

count

source

Web_Application 200

dtype: int64

DATA STATUS:

count

PIPELINE EXECUTION COMPLETED SUCCESSFULLY

Total Batches Processed: 20

Total Records Processed: 200

Cloud Storage Location:

/content/cloud_storage

Final Processed Dataset Saved:

/content/cloud_storage/final_pipeline_data.csv

FINAL DATA SAMPLE:

1 to 10 of 10 entries

index	timestamp	source	device_id	metric	value	page	hour	minute	status	normalized_value
0	2026-08-21 16:07:03.822232	Web_Application	USER_119	response_time	287.62	/checkout	16	7	NORMAL	0.
1	2026-08-21 16:07:03.822264	Web_Application	USER_106	response_time	451.46	/checkout	16	7	NORMAL	0.1329665067887257
2	2026-08-21 16:07:03.822271	Web_Application	USER_189	response_time	610.41	/home	16	7	NORMAL	0.2619644697652147
3	2026-08-21 16:07:03.822276	Web_Application	USER_129	response_time	333.77	/home	16	7	NORMAL	0.0374536394549541
4	2026-08-21 16:07:03.822279	Web_Application	USER_125	response_time	411.76	/checkout	16	7	NORMAL	0.1007474496627954
5	2026-08-21 16:07:03.822282	Web_Application	USER_132	response_time	512.53	/checkout	16	7	NORMAL	0.1825286684683368
6	2026-08-21 16:07:03.822289	Web_Application	USER_117	response_time	961.3	/home	16	7	NORMAL	0.546733864095634
7	2026-08-21 16:07:03.822292	Web_Application	USER_192	response_time	1519.81	/checkout	16	7	SLOW	1.
8	2026-08-21 16:07:03.822295	Web_Application	USER_196	response_time	1406.03	/home	16	7	NORMAL	0.907660344589714
9	2026-08-21 16:07:03.822298	Web_Application	USER_174	response_time	932.66	/login	16	7	NORMAL	0.523490695428464

Techniques Used

- Stream Processing
- Distributed Messaging using Apache Kafka
- Big Data Pipeline Design
- Data Serialization
- Apache Spark Streaming
- Distributed Storage using HDFS/S3
- Real-Time Data Visualization
- 5Vs of Big Data: Volume, Velocity, Variety, Veracity, and Value

Applications

- IoT and Smart City Data Processing
- Real-Time Financial Transaction Monitoring
- Fraud Detection Systems
- Social Media Analytics
- Healthcare Monitoring Systems
- Industrial and Manufacturing Monitoring
- E-Commerce Recommendation Systems



Future Scope

- Integrate more IoT and enterprise data sources.
- Implement advanced anomaly detection.
- Add machine-learning-based real-time prediction.
- Use containerization with Docker and Kubernetes.
- Implement automatic scaling.

Conclusion

- Provides real-time ingestion of distributed data.
- Kafka enables reliable data streaming.
- Spark Streaming enables fast data processing.
- HDFS/S3 provides scalable data storage.
- Grafana/Kibana provides real-time visualization.
- The pipeline supports reliable and low-latency analytics.

References

- Pacella, Massimo, et al. "A scalable framework for sensor data ingestion and real-time processing in cloud manufacturing." *Algorithms* 18.1 (2025): 22.
- Pacella, Massimo, Antonio Papa, Gabriele Papadia, and Emiliano Fedeli. "A scalable framework for sensor data ingestion and real-time processing in cloud manufacturing." *Algorithms* 18, no. 1 (2025): 22.
- Pacella M, Papa A, Papadia G, Fedeli E. A scalable framework for sensor data ingestion and real-time processing in cloud manufacturing. *Algorithms*. 2025 Jan 4;18(1):22.
- Pacella, M., Papa, A., Papadia, G., & Fedeli, E. (2025). A scalable framework for sensor data ingestion and real-time processing in cloud manufacturing. *Algorithms*, 18(1), 22.

**THANK
YOU**